

The Reader and the Frame

The Annihilator Lattice of the Digit-Extraction Towers, its Instrument-Dependence, and the Condition that Makes the Classical Formula Unique

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August, 2026

ABSTRACT

Two prior papers in this programme characterise a producer: the coefficient vector that, applied to the shell sums of a frame (b, n) , returns π or $\pi/\sqrt{3}$. This paper characterises the other half of the same operator — the annihilator lattice K of vectors that return zero — and finds that it does not behave like the producer at all.

The producer has a frame-intrinsic characterisation. The annihilator does not. We prove that $\dim K$ is a readout of the frame through one chosen coefficient field rather than a property of the frame, that the frame-intrinsic quantity is instead the multiplicative relation lattice R of the numbers $1 - 1/\alpha$ over the roots of $x^N - b$, and that $\dim K$ over the algebraic closure equals $\dim R$ exactly. The gap between the rational reading and the ceiling is not small: the frame $(16, 12)$, carried through the prior corpus as the empty holdout that made a lineage predicate falsifiable, is not empty. Its relation is exhibited over $\mathbb{Q}(2^{1/3})$ with residual 5.1×10^{-260} . The frame was never flat; the reader was rational.

The two halves are then locked together by a single number. The zero-sum condition C_1 that selects the producer is a single linear functional Σ , and Σ is non-zero on the annihilator: $\Sigma = -15$ on the octal kernel, -21 on the hexagonal. So the producer is unique exactly when Σ is injective on K , which for $\dim K = 1$ it is, and for $\dim K > 1$ it is not. Measured across the eight towers of the prior classification, uniqueness holds at seven and fails at exactly one — $(4096, 24)$, where C_1 leaves four dimensions unfixed. That is the $n = 24$ index collision the prior paper singles out for an unrelated reason. Because $\dim K$ is reader-dependent, producer uniqueness inherits that dependence, and this had not been stated.

Both classical seed kernels are then derived rather than searched, with no integer-relation detection, no coefficient bound and no precision threshold, so none of the instrument failures catalogued in §13 can reach them. The frame's odd-prime content is identified as the factorisation of $b - 1$, from which an exclusion sieve rigorously eliminates 1952 of 2233 frames with zero false exclusions. A grading theorem determines exactly

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when class-pure detection is complete. Winding is corrected from identity to strict inclusion, ancestry is closed over $\pm b^{1/k}$, and a third annihilator seed is found in the half of that family a guard clause had made unreachable. Section 13 lists everything killed, including six instrument failures and five claims originated and killed within this work, because the pattern is itself the finding: every forward claim died and every backward derivation lived.

1. The other half of the operator

A prior paper in this programme, *Objects as Closures*, asks where the coefficients of the Bailey–Borwein–Plouffe formula live, and answers that they are produced at a closure rather than possessed by either constraint. A second, *Exactly Two Seeds*, classifies the family that closure generates: $\sec^m(2\pi/m)$ is an admissible integer for exactly $m \in \{6, 8\}$, the generator is $4S(1) - 2S(m-4) - S(m-3) - S(m-2)$, and eight towers follow by lifting. Both papers are about a vector that returns a constant.

There is a second object in the same construction, and it is not the same object. For a frame (b, N) define the shells

$$S_r = \sum_{k \geq 0} 1 / ((Nk + r) b^k), \quad r = 1, \dots, N-1$$

A producer is a vector c with $\sum_r c_r S_r$ equal to a target constant. An annihilator is a vector with $\sum_r c_r S_r = 0$. The annihilators form a lattice, called here the kernel K , and essentially every published statement about relation spaces in this family is a statement about $\dim K$. The producer is a point in an affine fibre; the annihilator lattice is the fibre's direction space. They are complementary, not alternative, and the relationship between them turns out to carry the whole result.

1.1 A correction, first

An earlier working draft of this material identified the octal generator of K as “the Bailey–Borwein–Plouffe formula.” That is wrong, and the error is worth exhibiting rather than silently repairing, because it is the error the whole paper is organised around. Run both vectors against the shells:

```
=== octal (16,8), shells r = 1..7 ===
[8,-8,-4,-8,-2,-2,1] -> -9.681479787123295682045077e-122 SUM = -15
[4, 0, 0,-2,-1,-1,0] -> 3.141592653589793238462643 SUM = 0
pi = 3.141592653589793238462643

=== hexagonal (6,6), shells r = 1..5 ===
[16,-24,-8,-6,1] -> 4.598702898883565448971411e-121 SUM = -21
[4,-2,-1,-1,0] -> 2.418399152312290467458771 SUM = 0
(4/3)*pi/sqrt3 = 2.418399152312290467458771

producer proportional to kernel? octal: False hex: False
```

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The first vector in each pair annihilates; the second produces. They are linearly independent, and the classical formula is the second. What separates them is visible in the last column: the annihilator has non-zero coefficient sum and the producer has zero. That is not incidental. It is C_1 , the zero-sum or DC-blocking condition of the prior papers, and §6 shows it is precisely what makes the producer unique.

1.2 What the split buys

Separating the two halves resolves a naming collision that has caused real confusion in this corpus. The phrase “exactly two seeds” has been used for two different theorems about two different objects. In Exactly Two Seeds it is a proved statement about producers: the primitive angle $2\pi/m$ admits an integer radix for $m \in \{6, 8\}$ only, by an obstruction in field theory. In the relation-lattice work it was an unproved sweep result about annihilators: over $b \in [2, 320]$, $n \in [3, 9]$, only two frames carry a non-inherited kernel. The first survives everything in this paper. The second does not — §9 exhibits a third — and no contradiction arises, because they are claims about different halves of the operator.

Claimed as new in this paper: (1) the reduction of K to an S -unit relation lattice and a derivation of both classical seed kernels with no integer-relation search; (2) the identification of the frame's odd-prime content as the factorisation of $b - 1$, and a rigorous exclusion sieve built from it; (3) a grading theorem determining exactly when class-pure detection is complete; (4) the correction of winding from identity to inclusion, of ancestry to a closed family over $\pm b^{1/k}$, and a third annihilator seed found in the previously unsearched half; (5) the reader ladder, $\dim R$ as the invariant, and the demonstration that a headline zero in the corpus was an artefact of the coefficient field; and (6) the uniqueness bridge of §6, which is the only result here that speaks to both halves at once.

2. The problem, and its owner

The closed form on which everything here depends is not new. Borwein, Borwein and Galway, Finding and Excluding b -ary Machin-Type Individual Digit Formulae, Canadian Journal of Mathematics 56 (2004) 897–925, give at their equation (64)

$$L(1, b, n, j) = -(1/n) b^{(j/n)} \sum_r \zeta_n^{rj} (-rj) \ln(1 - \zeta_n^r b^{(-1/n)})$$

Substituting $\alpha = b^{1/n} \zeta_n^{-r}$ turns this into $S_r = -(1/n) \sum_j \alpha_j^r \log(1 - 1/\alpha_j)$, which is the identity this programme has been calling “the closed form.” Verified to 120 digits on (16, 8), (64, 6), (4096, 12) and (4096, 24). It is theirs and must be cited as theirs. Their valuation-vector method (§3.2) is likewise the origin of what this programme has called $\text{corank}(V_{\text{val}})$, and they themselves flag its blind spot: knowledge of the p -adic valuations for all p determines the imaginary part of the logarithm only modulo 2π , so the method returns potential dependencies. Every phase phenomenon reported in this corpus lives in that acknowledged gap.

Their closing Research Problem 10 asks whether their techniques can determine when a constant is a linear combination of $L(1, b^m, n, j)$ with b , and perhaps n , fixed, and remarks that this would probably require a good understanding of $\mathbb{Q}[\zeta_n, b^{1/n}]$. That is this paper's problem, stated by its owners in 2004, together with a correct diagnosis of what is missing. This paper supplies part of the missing understanding of that field and

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reports what it buys. It does not claim to answer Problem 10. Research Problem 2 asks how many $\mathbb{Q}$ -linearly independent binary Machin-type zero relations exist — that is $\dim K$, and §10 argues the question as posed is underdetermined without naming a reader.

3. Setup and conventions

Fix b and N . Let $\theta = b^{1/N}$ (real positive branch for $b > 0$), $\omega = \zeta_N$, and $\alpha_j = \theta\omega^j$ for $j = 0, \dots, N-1$, so that the α_j are the roots of $x^N - b$. Put $\beta_j = 1 - 1/\alpha_j$ and $L_j = \log \beta_j$ on the principal branch. For $c \in \mathbb{Q}^{(N-1)}$ write $C(y) = \sum_r c_r y^r$; note $C(0) = 0$, since the $r = 0$ shell diverges and is excluded.

Factor $x^N - b = \prod_i g_i$ over $\mathbb{Z}$. For monic g of degree d , $\prod(1 - \alpha) = g(1)$ and $\prod \alpha = (-1)^d g(0)$, so the orbit norm is $g(1)/g(0)$. The two classical frames:

seed	frame	orbits of $x^N - b$	orbit norms	kernel generator	Σ
octal	(16, 8)	$x^2-2, x^2+2, x^2-2x+2, x^2+2x+2$	$1/2, 3/2, 1/2, 5/2$	$[8, -8, -4, -8, -2, -2, 1]$	-15
hexagonal	(64, 6)	$x-2, x+2, x^2-2x+4, x^2+2x+4$	$1/2, 3/2, 3/4, 7/4$	$[16, -24, -8, -6, 1]$	-21

The corresponding producers are $[4, 0, 0, -2, -1, -1, 0]$ on the octal frame — the Bailey–Borwein–Plouffe formula for π , cite Bailey–Borwein–Plouffe 1997 — and $[4, -2, -1, -1, 0]$ on the hexagonal, returning $(4/3) \cdot \pi/\sqrt{3}$. Both have coefficient sum zero. Neither is a kernel element.

4. The reduction: K is an S -unit relation lattice

Using the closed form, $\sum_r c_r S_r = 0$ becomes $\sum_j C(\alpha_j) L_j = 0$. The $C(\alpha_j)$ are algebraic and the L_j are logarithms of algebraic numbers. Baker's theorem — if $\alpha_1, \dots, \alpha_n$ are non-zero algebraic with $\log \alpha_1, \dots, \log \alpha_n$ linearly independent over $\mathbb{Q}$, then $1, \log \alpha_1, \dots, \log \alpha_n$ are linearly independent over the algebraic closure — forces the coefficient vector into $R \otimes \mathbb{Q}$, where $R \subset \mathbb{Q}^N$ is the $\mathbb{Q}$ -space of relations among the L_j . Hence

$$K = \{c : (C(\alpha_j))_j \in R \otimes \mathbb{Q}\}$$

and R is a multiplicative relation lattice among the β_j inside the S -unit group of $\mathbb{Q}(\zeta_N, \theta)$ — the field BBG named. R is cut out by two independent families of conditions: valuation conditions (V), one per prime ideal in S , which is BBG §3.2; and argument conditions (A), from the unit and root-of-unity part, invisible to (V) by BBG's own caveat. (V) alone overcounts. Both together give R exactly.

4.1 Two immediate consequences

Lemma 4.1 (sum-zero). Every $c \in R$ satisfies $\sum_j c_j = 0$.

Proof. Let β divide b . Then $v_\beta(\alpha_j) = v_\beta(b)/N$ is the same for every j , since the ω^j are units, and $v_\beta(\alpha_j - 1) = 0$, so $v_\beta(\beta_j) = -v_\beta(\alpha_j)$ is independent of j . The valuation condition at β reads $-v_\beta \cdot \sum_j c_j = 0$. $\square$

Verified on every relation vector computed in this work. Note that this is a condition on R , the log side. It is not the zero-sum condition C_1 on the coefficient side, which §6 shows is violated by every kernel generator measured here.

Theorem 4.2. $\dim K$ over the algebraic closure equals $\dim R$.

Proof. Over $\mathbb{Q}$ the map $c \mapsto (C(\alpha_j))_j$ from $\mathbb{Q}^{N-1}$ to $\mathbb{Q}^N$ is injective, with image the span of the columns $(\alpha_j^r)_j$ for $r = 1, \dots, N-1$. Its one-dimensional annihilator is spanned by $(1, \dots, 1)$, because the power sums $\sum_j \alpha_j^r$ vanish for $1 \leq r \leq N-1$. So the image is exactly the sum-zero hyperplane, which contains $R \otimes \mathbb{Q}$ by Lemma 4.1. Hence the intersection is all of $R \otimes \mathbb{Q}$. $\square$

Together with Baker, this says the ladder of readers terminates at $\dim R$ and cannot exceed it. $\dim R$ is the frame-intrinsic quantity; $\dim K$ is one rung.

5. The closure on the annihilator side

The prior papers select the producer by imposing two independently motivated constraints simultaneously and reading what neither can move. The same architecture operates on the annihilator side, with different constraints, and it is worth setting out in the same bookkeeping because it is the reason both seed kernels can be derived without any search.

Forcing lemma. If a prime ideal θ divides exactly one β_j , then every $c \in R$ has $c_j = 0$; since R is Galois-stable, c vanishes on the whole orbit of α_j , so C is divisible by that orbit's minimal polynomial.

5.1 Octal, (16, 8)

$\theta = 16^{(1/8)} = \sqrt[4]{2} \in \mathbb{Q}(\zeta_8)$, so $\mathbb{Q}(\zeta_8, \theta) = \mathbb{Q}(\zeta_8)$, degree 4, unit rank 1 with fundamental unit $u = 1 + \sqrt[4]{2}$, and 2 totally ramified. With $s = \sqrt[4]{2}$ and $z = \zeta_8$: $\beta_0 = u^{-1}s^{-1}$, $\beta_4 = u s^{-1}$, $\beta_1 = z s^{-1}$, $\beta_7 = z^{-1}s^{-1}$, while β_2 and β_6 carry the two conjugate primes over 3 and β_3, β_5 the two over 5, each unshared. So $c_2 = c_3 = c_5 = c_6 = 0$, and on $\{0, 1, 4, 7\}$ the conditions give $c_0 = c_4$, $c_1 = c_7$, $\sum c = 0$:

$$R = \text{span}\{(1, -1, 0, 0, 1, 0, 0, -1)\}, \quad \dim R = 1$$

Therefore C is divisible by $y(y^2+2)(y^2+2y+2)$, leaving $C(y) = y(y^2+2)(y^2+2y+2)(a + by + cy^2)$ with $C(\alpha_0) = C(\alpha_4) = \lambda$ and $C(\alpha_1) = C(\alpha_7) = -\lambda$. Solving gives a single free parameter, hence $\dim K = 1$ forced, and $C = [8, -8, -4, -8, -2, -2, 1]$. Series residual 4.9×10^{-80} . No integer-relation detection, no coefficient bound, no precision threshold.

5.2 Hexagonal, (64, 6), and the vanishing counted

$\theta = 64^{(1/6)} = 2 \in \mathbb{Q}$, so the field is $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$, degree 2, unit rank 0 — units are torsion only, so (A) does real work here. With 2 inert, 3 ramified, and 7 split:

j	β_j	(v_2, v_3, v_7, v_7)	arg
0	$1/2$	$(-1, 0, 0, 0)$	0

j	β_j	$(v_2, v_3, v_7, \bar{v}_7)$	arg
1	$(3 + i\sqrt{3})/4$	$(-1, 1, 0, 0)$	$\pi/6$
2	$(5 + i\sqrt{3})/4$	$(-1, 0, 1, 0)$	irrational
3	$3/2$	$(-1, 2, 0, 0)$	0
4	$(5 - i\sqrt{3})/4$	$(-1, 0, 0, 1)$	irrational
5	$(3 - i\sqrt{3})/4$	$(-1, 1, 0, 0)$	$-\pi/6$

The dimensional bookkeeping is the argument, so it is set down in the form the prior papers use. (V) alone gives $\Sigma c = 0$, $c_1 + c_5 + 2c_3 = 0$, $c_2 + c_4 = 0$ — three dimensions, and nothing is selected. (A) alone gives $c_1 = c_5$, $c_2 = c_4$ — four dimensions, and nothing is selected. Together they give one dimension, generator $(-1, 1, 0, -1, 0, 1)$, residual 8.7×10^{-62} .

constraints imposed	admissible dimension	is anything selected?
none	6	no
(V) only — valuations	3	no
(A) only — arguments	4	no
(V) and (A) together	1	yes — a unique line

The two-dimensional gap between (V) alone and (V) with (A) is exactly the layer BBG say valuations cannot see. Proceeding as before, $C = y(y^2 + 2y + 4)(a + by + cy^2)$ with one free parameter, giving $\dim K = 1$ and $C = [16, -24, -8, -6, 1]$, series residual 4.1×10^{-80} .

The vanishing test of the prior papers applies unchanged: remove (V) and the selection is destroyed, remove (A) and the selection is destroyed. Neither reduction is doing part of the other's work — (V) removes three dimensions, (A) removes two, and together they remove five, exactly the sum. This is the same closure architecture the producer side uses, on the other half of the operator, with constraints that mention neither the producer nor any target.

5.3 Consequence for the corpus

The statement “the principal relation annihilates exactly the unshared-prime orbits,” previously carried as verified by measurement, is now a one-line corollary of the forcing lemma: an unshared prime has nothing to cancel against, so its coordinate must vanish. The count of such orbits is a rank deficiency, not an auxiliary statistic.

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6. Prime content is the factorisation of $b - 1$

Since $x^N - b = \prod_i g_i$, evaluation at 1 gives $\prod_i g_i(1) = 1 - b$. So the odd-prime content of a frame is the factorisation of $b - 1$, distributed by N and never created by it. The published winding table, presented as evidence that winding manufactures primes, is the factorisation of $b^k - 1$ — that is Cunningham and Zsigmondy, and BBG's own Theorem 3, after Bang, is the exclusion tool built from it.

Proposition 6.1. If $v_p(b - 1) = 1$ then p divides exactly one $g_i(1)$, to first power, and p is unshared.

frame	$b - 1$	unshared (published)	$\{p : v_p(b-1) = 1\}$
(16, 8)	$3 \cdot 5$	$\{3, 5\}$	$\{3, 5\}$
(256, 16)	$3 \cdot 5 \cdot 17$	$\{3, 5, 17\}$	$\{3, 5, 17\}$
(4096, 24)	$3^2 \cdot 5 \cdot 7 \cdot 13$	$\{5, 7, 13\}$	$\{5, 7, 13\}$
(65536, 32)	$3 \cdot 5 \cdot 17 \cdot 257$	$\{3, 5, 17, 257\}$	same
(1048576, 40)	$3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41$	$\{3, 11, 31, 41\}$	same
(64, 6)	$3^2 \cdot 7$	$\{7\}$	$\{7\}$
(4096, 12)	$3^2 \cdot 5 \cdot 7 \cdot 13$	$\{5, 7, 13\}$	same
(262144, 18)	$3^3 \cdot 7 \cdot 19 \cdot 73$	$\{7, 19, 73\}$	same
(16777216, 24)	$3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$	$\{5, 7, 13, 17, 241\}$	same

The rule that 3 moves in and out of shared status is simply whether g divides $b - 1$. The converse of Proposition 6.1 is false. Searching $n \in [3, 12]$, $b \in [2, 400]$ produced twenty counterexamples — primes sitting in exactly one orbit with multiplicity greater than one. Example: (64, 4), where $b - 1 = 63 = 3^2 \cdot 7$ and 3 occupies a single orbit at exponent 2. Also (100, 4), (289, 4), (361, 4), (243, 5), (100, 6), (64, 8). Only the implication survives, and every downstream use must take the safe direction.

6.1 The exclusion sieve

Lemma 6.2. If $v_p(b - 1) = 1$, exactly one prime ideal over p has non-zero valuation on the β 's; it has residue degree 1 and divides exactly one β_j ; so $c_j = 0$ and, by Galois-stability, C is divisible by g_i .

Corollary 6.3 (necessary for $\dim K > 0$). $1 + \sum \deg g_i$, over orbits carrying such a prime, must be at most $N - 1$, the leading 1 because y divides C .

Factorisation only — no integer-relation detection, no precision, no coefficient bound. Over $b \in [2, 320]$ and $N \in [3, 9]$: 2233 frames examined, 1952 rigorously excluded (87.4 percent), zero false exclusions against sixteen frames of known non-zero $\dim K$, and 281 survivors, of which 245 have $x^N - b$ irreducible and 36 reducible.

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Both classical seeds are among the 36. The sieve is necessary, not sufficient, and it is silent on (16, 12) — which §10 shows was the right silence for the wrong reason.

7. The uniqueness bridge

This section is the only one that speaks to both halves of the operator at once, and it is the reason the split of §1 is worth making.

7.1 Σ on K

Let Σ denote the coefficient-sum functional, $c \mapsto \Sigma_r c_r$. The prior papers impose C_1 : $\Sigma(c) = 0$, motivated as the requirement that the instrument have no response to a constant. Two producers differ by an annihilator, so the producer at a frame is unique modulo K , and C_1 pins it down exactly when Σ is injective on K — that is, when $\dim K$ minus the rank of Σ restricted to K is zero. Write that difference as the slack.

This is a sharp criterion and it is cheap to evaluate. Measured live, at 300 decimal digits with complete relation sets by recursive elimination:

frame	$\dim K$ over $\mathbb{Q}$	Σ on the kernel basis	slack	producer unique?
(16, 8)	1	[−15]	0	yes
(64, 6)	1	[−21]	0	yes
(16, 12)	0	[]	0	yes, vacuously over $\mathbb{Q}$
(16, 24)	3	[−15, −30, 45]	2	no — 2-dim slack
(4096, 12)	1	[1365]	0	yes
(4096, 24)	5	[0, −2730, 0, −1365, 10920]	4	no — 4-dim slack
(−64, 12)	2	[0, −65]	1	no — 1-dim slack
(256, 16)	1	[−255]	0	yes

Worst residual per frame across this table ranges from 7.9×10^{-301} to 1.5×10^{-297} , all far below the gate of 10^{-230} , and dimensions are stable across maxcoeff 10^5 to 10^7 where tested.

7.2 The eight towers

Exactly Two Seeds predicts and verifies eight towers. Its verification is of the value each returns, which is not in question here. What was not established is whether the coefficient vector at each is the unique DC-blocked vector, and that is exactly the slack question. Run on all eight:

family	k	n	b	dim K over $\mathbb{Q}$	slack	producer unique?
$45^\circ (m = 8)$	1	8	16	1	0	yes
$45^\circ (m = 8)$	2	16	256	1	0	yes
$45^\circ (m = 8)$	3	24	4096	5	4	NO
$45^\circ (m = 8)$	4	32	65536	1	0	yes
$60^\circ (m = 6)$	1	6	64	1	0	yes
$60^\circ (m = 6)$	2	12	4096	1	0	yes
$60^\circ (m = 6)$	3	18	262144	1	0	yes
$60^\circ (m = 6)$	4	24	16777216	≥ 1 (see below)	—	unmeasured

Uniqueness holds at six of the eight measured, fails at exactly one, and is unmeasured at one. The failure is at $(4096, 24)$, which is the frame Exactly Two Seeds §7 singles out as the $n = 24$ collision where the two families meet at different radices. It flags that frame for an index reason; it is anomalous here for an entirely different reason, and the coincidence is worth recording rather than explaining. At that frame C_1 removes one dimension from a five-dimensional annihilator lattice and four remain. The generator's vector is a producer there; it is not the only one.

Instrument failure, caught and corrected. The detector returned $\dim K = 0$ at $(16777216, 24)$. That cannot be right: $W \cdot K_{\text{seed}} \subseteq K_{\text{frame}}$ is established in §8, so the $k = 4$ hexagonal lift must be present. Constructing it rather than searching for it — $C(y) = W_4(y) \cdot D(y)$ with $W_4(y) = 64^3 + 64^2y^6 + 64y^{12} + y^{18}$ — gives a vector of height 6291456 and $\Sigma = -5592405$, with residual 2.93×10^{-295} against the shells. The detector's maxcoeff was 10^6 . The zero was a false negative produced by a coefficient bound, exactly the failure mode §13.3 catalogues, and it is reported here rather than in the table because a zero from a detector is not a zero.

7.3 $\Sigma(K)$ is divisible by the unshared content

The Σ column above is not arbitrary, and it need not be measured. $\Sigma(c) = C(1)$, and the forcing lemma already says what divides C : for every unshared-prime orbit, C is divisible by g_i . Hence $C(1)$ is divisible by the product of $g_i(1)$ over those orbits, and by Proposition 6.1 each such $g_i(1)$ carries its unshared prime once. Writing $U(b)$ for the product of the odd primes p with $v_p(b - 1) = 1$:

$$\Sigma(K) \subseteq U(b) \cdot \mathbb{Z}$$

Derived, not searched. Tested against the measured Σ lattices:

frame	$b - 1$	$U(b)$ predicted	$\gcd \Sigma(K)$ measured	ratio	divides?
(16, 8)	$3 \cdot 5$	15	15	1	yes
(64, 6)	$3^2 \cdot 7$	7	21	3	yes
(16, 24)	$3 \cdot 5$	15	15	1	yes
(4096, 12)	$3^2 \cdot 5 \cdot 7 \cdot 13$	455	1365	3	yes
(4096, 24)	$3^2 \cdot 5 \cdot 7 \cdot 13$	455	1365	3	yes
(-64, 12)	$5 \cdot 13$	65	65	1	yes
(256, 16)	$3 \cdot 5 \cdot 17$	255	255	1	yes

Seven of seven. The divisibility is a corollary and is claimed as proved. The exact generator is not: the ratio column is 1 or 3, and it is 3 exactly where g divides $b - 1$ — which is §6's own rule for when the prime 3 is shared. That correlation is measured on seven frames and is not derived; it is tagged $\circ$ unproven in the ledger. The immediate consequence that is derived: a frame with empty unshared content cannot have Σ non-zero on K by this route, so producer uniqueness and prime content are not independent questions.

8. The θ -grading theorem, and Δ -phase

Let $d = [\mathbb{Q}(\zeta_n, \theta) : \mathbb{Q}(\zeta_n)]$. For $b = 2^m$ this is computable in closed form: write $m/N = p/g$ in lowest terms, so $\theta = 2^{p/g}$; then $d = g/2$ if g is even and 8 divides N , since $\sqrt{2} \in \mathbb{Q}(\zeta_n)$, and $d = g$ otherwise.

Theorem 8.1. For any u in the orthogonal complement of R with rational entries, $\sum_j u_j C(\alpha_j) = \sum_r c_r \theta^r T_r(u)$, where $T_r(u) = \sum_j u_j \omega^{(jr)} \in \mathbb{Q}(\zeta_n)$. Since $\{1, \theta, \dots, \theta^{d-1}\}$ is a $\mathbb{Q}(\zeta_n)$ -basis of $\mathbb{Q}(\zeta_n, \theta)$ and $\theta^d \in \mathbb{Q}(\zeta_n)$, the condition splits into d independent conditions indexed by $r \bmod d$. Hence K decomposes as a direct sum over the residues mod d .

Corollary 8.2. When d equals the subdivision index q , every kernel element is class-pure and the class-pure detector is complete, not a probe. When $d < q$ the grading is coarser than the class structure, nothing forces purity, and mixed relations may exist that a class-pure detector cannot see.

8.1 Winding raises the period

A parameter error propagated through the corpus and inverted several readings. Winding maps (b, n) to (b^m, nm) — it raises the period as well as the base, which is exactly why $x^{(nm)} - b^m$ contains $x^n - b$ as a factor. Therefore (4096, 24) is the octal seed wound $k = 3$, with $q = 1$. It is not a $q = 3$ subdivision, and no character or carrier apparatus applies to it. Likewise (256, 16) is octal wound $k = 2$ with $q = 1$, and (4096, 12) is hexagonal wound $k = 2$ with $q = 1$. Measured, with all parameters corrected:

frame	n	q	d	dim K	sum of class dims	gap
(16, 16)	8	2	2	1	1	0
(16, 24)	8	3	3	3	3	0
(16, 32)	8	4	4	1	1	0
(64, 24)	6	4	2	1	1	0
(4096, 12)	6	1	1	1	0	1
(4096, 24)	8	1	1	5	0	5

8.2 Δ_{phase} does not exist

The corpus reports $\Delta_{\text{phase}} = \dim K - \text{corank}(V_{\text{val}}) = 2$ at exactly (16, 24) and (4096, 24), and treats it as the central mystery. At (16, 24), $\dim K = 3$, not 5. Two independent methods agree: the full 23-shell iterated detection returns exactly three relations, stable from maxcoeff 10^5 to 10^7 at residual of order 10^{-301} , with supports and heights

```
maxcoeff 1e5, 1e6, 1e7: dim K over Q = 3
supp [3, 6, 9, 12, 15, 18, 21]  r mod 3 -> [0]  h=8  resid 1.28e-301
supp [1, 4, 7, 10, 13, 16, 19, 22]  r mod 3 -> [1]  h=32  resid 2.62e-301
supp [2, 5, 8, 11, 14, 17, 20, 23]  r mod 3 -> [2]  h=48  resid 3.50e-301
```

and the class-by-class decomposition returns $1 + 1 + 1$. All three are class-pure, as Theorem 8.1 requires since $d = q = 3$. So $\Delta_{\text{phase}} = 0$ there and cannot be otherwise. At (4096, 24) the gap is real but it is 5, not 2: none of the five relations is class-pure, because $d = 1$ and nothing forces purity. The carriers previously reported at that frame were read through the winding map, not measured in the frame — the same word for a different object.

Δ_{phase} is therefore not $C - 1$, not $\gcd(C, q) - 1$, and not a phase correction. It is the failure of the θ -grading, and it is provably zero whenever $d = q$. One of its two data points was a detector error; the other belongs to a family that also contains (4096, 12).

9. Winding is transport, not generation

The corpus states $D_{\text{frame}}(c) = W(x) D_{\text{seed}}(c)$ as an identity and calls the three-layer map strictly invertible. Its own record contradicts this: at (4096, 72) the class-0 relation is not divisible by W . The correct statement is an inclusion. With $x^{24} - 4096 = (x^8 - 16)W$ and $W = y^{16} + 16y^8 + 256$, the product $W \cdot D_{\text{seed}}$ has residual 1.52×10^{-298} against the shells of (4096, 24), so $W \cdot K_{\text{seed}} \subseteq K_{\text{frame}}$ holds; but $\dim K_{\text{frame}} = 5$ while $\dim W \cdot K_{\text{seed}} = 1$.

$$W \cdot K_{\text{seed}} \subsetneq K_{\text{frame}}$$

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Strictly. “Algebraically independent layers” and “strictly invertible” must come out; what is demonstrated is that the two operations act on distinct coordinates of the construction.

9.1 Lift heights are deterministic

The lift is polynomial multiplication, so heights multiply exactly. Measured live:

lift	source height	$ W _{\infty}$	product	measured	Σ of lift
octal into (256, 16), $k = 2$	8	16	128	128	—
octal into (4096, 24), $k = 3$	8	256	2048	2048	-4095
hex into (4096, 12), $k = 2$	24	64	1536	1536	-1365
hex into (4096, 24), $q = 2, k = 2$	24	64	1536	1536	—
hex into (16777216, 24), $k = 4$	24	262144	6291456	6291456	-5592405
octal into (16, *), q only	8	1	8	8	—

Height therefore tracks inheritance depth, not any property of the frame. A conjecture that height measures how underconstrained a frame is died on measurement: the shortest vector in $K(4096, 24)$ has height 256, below both inherited lifts. The 27648 appearing in the raw basis is an artefact of basis choice.

10. Ancestry, and a third annihilator seed

10.1 Ordering

The corpus computes native structure as $\dim K_{\text{frame}}$ minus the dimension of ancestral lifts and names the remainder an object. That number is a statement about the enumerator until the enumerator is proved complete. In this work it moved from 4 to 3 to 1 at a single frame without any mathematics changing. The correct order is representation completeness, then ancestry completeness, then native.

10.2 The ancestor family, closed

A frame (b_0, m) feeds (b, N) when $b_0^k = b$ and $m^k = N$, with lift $C(y) = W_k(y) \cdot D(y^q)$ and $W_k(y) = \sum_{i < k} b_0^{(k-1-i)} y^{(mqi)}$, subdivision first then winding. The real integer solutions of $b_0^k = b$ are $b_0 = \pm b^{(1/k)}$, negative only for even k . That family is finite, enumerable, and complete within the class of frames $x^m - b_0$ with $b_0 \in \mathbb{Z}$.

The enumerator used to produce the corpus contained a guard, $r \geq 2$, in its integer-root routine. The entire negative half of the family was structurally unreachable. It was not a mathematical gap; it was an authored one.

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10.3 Consequences

$4096 = (-64)^2$, and $x^{24} - 4096 = (x^{12} - 64)(x^{12} + 64)$ — the right-hand factor is the frame $(-64, 12)$, present in the factorisation the whole time. $K(-64, 12)$ has dimension 2, measured live:

$[0, 16, -16, -24, 0, 8, 4, 6, 6, 1, -1]$ height 24 Sigma = 0 residual $5.48e-301$

$[32, -32, -32, 0, -8, -16, -4, 0, -4, -2, 1]$ height 32 Sigma = -65 residual $5.72e-301$

Both lift into $K(4096, 24)$. Recomputing ancestry with the negative half restored moves the inherited dimension from 2 to 4 and native from 3 to 1. $(-64, 12)$ is a seed of the annihilator lattice: its ancestors are (b_0, m) with $b_0^k = -64$, so k odd — $(-64, 6)$, $(-64, 4)$, $(-64, 3)$, $(-64, 2)$ via $k = 1$, and $(-4, 4)$, $(-4, 2)$ via $k = 3$ — and every one has $\dim K = 0$. Both of its relations are native to it. It carries dimension 2 where both classical seeds carry 1, and its coefficient heights, 24 and 32, are smaller than the octal seed's.

10.4 Reconciliation with the seed theorem

This does not contradict Exactly Two Seeds, and the reason is the split of §1. That paper's theorem is that $\sec^m(2\pi/m)$ is an integer greater than one for exactly $m \in \{6, 8\}$, proved by an obstruction in field theory: $\mathbb{Q}(2\cos(2\pi/m))$ is abelian while $\mathbb{Q}(2^{1/e})$ is not for $e \geq 3$. It is a statement about admissible positive radices for producers. The sweep that produced “exactly two seeds” on the annihilator side ran $b \in [2, 320]$, $n \in [3, 9]$; $(-64, 12)$ is outside on both axes, and it is a claim about a different object.

The two are nonetheless adjacent at exactly one point, and it is worth marking. Exactly Two Seeds §6 remarks that at $m = 3$ the value $\sec^3(120^\circ) = -8$ is an integer, excluded by positivity rather than integrality. The negative half of the radix domain is precisely where that exclusion operates, and it is precisely where the annihilator sweep was never run. The two papers set aside the same region for two independent reasons, and it turned out to contain something. We claim no mechanism connecting them; we note that a third seed and a positivity exclusion sit on the same boundary, and that this is a place to look rather than a result.

10.5 The inheritance table

frame	dim K	inherited	native	h_min	inherited from
(16, 16)	1	1	0	8	(16, 8) $q = 2$
(16, 24)	3	1	2	8	(16, 8) $q = 3$
(16, 32)	1	1	0	8	(16, 8) $q = 4$; (16, 16) $q = 2$
(256, 16)	1	1	0	128	(16, 8) $k = 2$
(4096, 24)	5	$2 \rightarrow 4$	$3 \rightarrow 1$	256	(16, 8) $k = 3$; (64, 6) $q = 2, k = 2$; $(-64, 12)$ $k = 2$
(64, 12)	1	1	0	24	(64, 6) $q = 2$
(64, 18)	1	1	0	24	(64, 6) $q = 3$

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frame	dim K	inherited	native	h_min	inherited from
(64, 24)	1	1	0	24	(64, 6) q = 4; (64, 12) q = 2
(4096, 12)	1	1	0	1536	(64, 6) k = 2

Native is non-zero at exactly (16, 24) and (4096, 24) — the same two frames the corpus flagged for Δ _phase. The anomaly was correctly located and incorrectly measured. But the two are different phenomena: at (16, 24) native = 2 and those two are the class-1 and class-2 twisted carriers; at (4096, 24), q = 1 and no character twist exists at all. Negative-base ancestors were also enumerated for (16, 24) — eight candidates, all with dim K = 0 — so native stays 2 there. The two anomalous frames separate cleanly under a representation change: one survives it, one mostly does not.

11. The reader ladder

11.1 (16, 12) is not flat

The holdout frame reads dim K over $\mathbb{Q}$ = 0. But dim R = 1: the relation $-2L_0 + 3L_2 + L_4 - 6L_6 + L_8 + 3L_{10} = 0$ holds among the logs with residual 4.5×10^{-221} . Solving for a realising polynomial over $\mathbb{R}$ returns coefficients $8 \cdot 2^{1/3}$ and $2 \cdot 2^{1/3}$ — real, algebraic and irrational. Over $\mathbb{Q}(\theta)$ with $\theta = 16^{1/12} = 2^{1/3}$:

$$89S_1 - 16S_2 - 89S_4 + 4S_5 + 29S_7 - 4S_8 - 29S_{10} + S_{11} = 0, \text{ residual } 5.1 \times 10^{-260}$$

The frame is full. The reader was rational. This retires the ledger line “lineage predicate predicts absence: (16, 12) has dim K = 0” as evidence. dim K = 0 there is an instrument fact, and §7’s entry “producer unique, vacuously” at that frame inherits the same qualification: it is unique over $\mathbb{Q}$ and not over $\mathbb{Q}(\theta)$.

11.2 The invariant, and its ceiling

By Baker, dim R is unchanged by any widening of the coefficient field on the log side; by Theorem 4.2, dim K over the algebraic closure equals dim R. So the readings are monotone up a ladder that terminates at dim R. Measured rungs:

frame	$\mathbb{Q}$	$\mathbb{Q}(\theta)$	$\mathbb{Q}(\theta, \sqrt{3})$	dim R
(64, 6)	1	1	—	1
(16, 8)	1	1	—	1
(16, 12)	0	1	—	1
(16, 24)	3	3	5	5
(4096, 24)	5	—	—	7

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The (16, 12) row is the one that forces the reframe. That frame is carried throughout the prior corpus as the holdout — the rich-orbit frame with $\dim K = 0$, whose emptiness made the lineage predicate falsifiable. It is not empty.

11.3 The missing rung is cyclotomic, not radical

At (16, 24), widening by the radical buys nothing: $\mathbb{Q}(\theta)$ still reads 3. Both antisymmetric directions have real coefficients that are not in $\mathbb{Q}(2^{1/6})$ but are in $\mathbb{Q}(2^{1/6}, \sqrt{3})$. The mechanism: the antisymmetric part is built from $\sin(2\pi j r/24)$, which lies in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Since $\sqrt{2} = \theta^3$ already, the only element missing from $\mathbb{Q}(\theta)$ is $\sqrt{3} = \zeta_{12} + \zeta_{12}^{-1}$ — which comes from the roots of unity, not from $b^{1/N}$. The reader that attains the ceiling needs both sources at once; either alone stalls at 3.

12. Two channels, and why the law is not clean

Complex conjugation $\sigma : j \mapsto N - j$ acts on R ; write $R = R^+ \oplus R^-$ for the symmetric and antisymmetric parts. Since $\operatorname{Re} L_j$ is σ -symmetric and $\operatorname{Im} L_j$ antisymmetric, the two conditions decouple exactly, which is also how R is computed here. R^+ is the modulus channel, R^- the argument channel.

frame	θ	$\dim R$	R^+	R^-	$\dim K \text{ over } \mathbb{Q}$
(16, 24)	$2^{1/6}$	5	3	2	3
(4096, 24)	$\sqrt{2}$	7	3	4	5

At (16, 24), $\dim K \text{ over } \mathbb{Q}$ equals $\dim R^+$: the rational reader sees the modulus channel exactly and none of the argument channel. That suggested a law. The law is false. At (4096, 24), $\dim K \text{ over } \mathbb{Q}$ is 5 while $\dim R^+$ is 3, so $\mathbb{Q}$ reads two dimensions of the argument channel. The mechanism is visible in the θ column — $\theta = \sqrt{2} \in \mathbb{Q}(\zeta_{24})$, so the radical has collapsed into the cyclotomic field and $\mathbb{Q}$ inherits argument structure for free. At (16, 24), $\theta = 2^{1/6}$ sits outside, the sources stay separate, and $\mathbb{Q}$ sees only the modulus half. Reader-to-channel is not a partition. It is an overlap that depends on whether $d = 1$.

Two further facts about (4096, 24) worth recording. First, $\dim R = 7$, so even the rational reader's 5 is not the ceiling: two dimensions sit above every measurement in the corpus. Second, $4096 = 16^3 = 64^2$ — the frame is a confluence where the octal and hexagonal lineages meet, and it is also the frame with the largest relation space measured, with the excess concentrated in the argument channel, 4 against 2 for its sibling. It is the same frame at which §7 finds producer uniqueness failing. The coupling condition here is exact and arithmetic — a double perfect power with matching period division — not a threshold.

13. Reader reach is two-sided

The realisation map is explicit: given $w \in R$, the realising polynomial is unique up to scale by inverse discrete Fourier transform, $c_r = \lambda \hat{w}_r / (N\theta^r)$ with $\hat{w}_r = \sum_j w_j \omega^{(-jr)}$. Since this is linear in w , the set of directions a reader F can realise is a kernel, not a search. That converts “which reader sees what” from a per-field measurement into a derivation, and it was run.

Calibrated first: the instrument returns $\dim R = 1, 1, 1, 5$ on $(16, 8), (64, 6), (16, 12), (16, 24)$ with $R^+ = 3$ and $R^- = 2$ at the last, matching §12. Then the criterion, computed as the rational nullity of the conditions that the reader cannot see a component:

```
dim K over Q      = 1    measured: 3
dim K over Q(theta) = 3    measured: 3
dim K over Q(theta,sqrt3) = 5    measured: 5
```

Two of three rungs match. The rational rung does not, and the direct measurement of §8.2 — three class-pure relations, stable across three coefficient bounds at residual 10^{-301} — is the one that stands. The criterion is wrong, and where it is wrong is locatable. Pushing each measured kernel vector forward through $v_j = C(\alpha_j)$:

```
class 0: v/v[0] rational and real      <- found by the criterion
class 1: ratios at (1/2) z3^-1, (3/2) z3      <- missed
class 2: ratios at (1/2) z3 , (3/2) z3^-1      <- missed
z3 = primitive cube root of unity
```

The criterion restricted the R -side vector to $R \otimes \mathbb{Q}$. The twisted classes correspond to $w \in R \otimes \mathbb{Q}(\zeta_3)$, so only the untwisted direction was visible and the reading came back 1 instead of 3. The consequence is that a reader is a pair of fields, not one: a coefficient field on the c -side and a twist field on the R -side. Both of the clean laws proposed in §12 broke because they were single-sided statements about a two-sided object, and Open Problem 6 in §15, which asks when a reader attains the ceiling, names only the c -side and is therefore underdetermined as posed.

This reframes that problem; it does not answer it. It is logged in §14.2 as a claim originated and killed within this work.

14. Falsification ledger

Everything killed, with attribution. Instrument failures are listed as such because the pattern is the finding.

14.1 Corpus claims killed by measurement

claim	status	evidence
$\dim K(16, 24) = 5$	$\perp 3$	two independent methods, stable maxcoeff 10^5 – 10^7

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claim	status	evidence
$\Delta_{\text{phase}} = C - 1 = 2$ at exactly two frames	$\perp$	quantity does not exist; $= 0$ at $(16, 24)$ by Thm 8.1, $= 5$ at $(4096, 24)$
$D_{\text{frame}} = W \cdot D_{\text{seed}}$ identity	$\perp$	inclusion strict, quotient dim 4 before negative bases, 1 after
"strictly invertible" three-layer map	$\perp$	inverse undefined on the quotient
exactly two seeds (annihilator sweep)	$\perp$	$(-64, 12)$, $\dim K = 2$, all ancestors empty
$(4096, 24)$ is a $q = 3$ subdivision	$\perp$	winding raises the period; $q = 1$
$(16, 12)$ has no relation	$\perp$	relation exhibited over $\mathbb{Q}(2^{1/3})$, residual 5.1×10^{-260}
$\text{unshared} \Leftrightarrow v_p(b-1) = 1$	$\perp$ converse	20 counterexamples, e.g. $(64, 4)$
kernel generator is the BBP formula	$\perp$	it annihilates; $\Sigma = -15 \neq 0$; BBP is the producer, $\Sigma = 0$
$\dim K(16777216, 24) = 0$	$\perp$	false negative at maxcoeff 10^6 ; lift constructed, height 6291456
$N_{\text{active}} = \gcd(C, q)$	o unproven	proposed as a repair, never derived, never tested
carrier iff $d \mid C$	o unproven	no theorem connecting character order to carrier existence
$\Sigma(K)$ generator is $U(b)$ or $3U(b)$ by $9 \mid b-1$	o unproven	measured 7/7; divisibility is proved, the exact generator is not
class-0 at $(4096, 72)$ not divisible by W	o unmeasured	carried from the corpus, not reproduced here

14.2 Claims originated and killed in this work

claim	how it died
both seeds are exactly the Kummer-degenerate frames $b^{1/n} \in \mathbb{Q}(\zeta_n)$	every perfect n -th power in the sweep collapses the field and carries nothing: $(8, 3), (27, 3), (64, 3), (125, 3), (216, 3), (16, 4), (256, 4), (32, 5), (128, 7), (256, 8)$
height measures how underconstrained a frame is	$h_{\min}(4096, 24) = 256$, below both inherited lifts
$\dim K \text{ over } \mathbb{Q}(\theta) = \dim R$	true on $(64, 6), (16, 8), (16, 12)$ because $R^- = 0$ there; false at $(16, 24)$

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claim	how it died
$\dim K \text{ over } \mathbb{Q} = \dim R^*$	false at (4096, 24): 5 against 3
native = 3 at (4096, 24)	1, after negative-base ancestors
reader reach is determined by the coefficient field alone	reproduces $\mathbb{Q}(\theta)$ and $\mathbb{Q}(\theta, \sqrt{3})$ and fails at $\mathbb{Q}, 1$ against 3; the twisted classes need $R \otimes \mathbb{Q}(\zeta_3)$

14.3 Instrument failures

Every one of these produced a plausible number, not an error.

failure	effect	fix
real nullity computed where rational nullity required	over-reported $\dim K$; would have confirmed a false 1 at (16, 12)	grade by θ -powers, or test rationality of the null vector
$\lambda = 1$ fixed in the interpolation	reported 2 of 5 R-directions unrealisable at (16, 24)	free phase; antisymmetric directions need $i \cdot C$
$r \geq 2$ guard in the integer-root routine	entire negative ancestor half invisible; native inflated $1 \rightarrow 3$	$b_0 = \pm b^{1/k}$
int() applied to rational nullspace vectors	silent truncation	clear denominators
LLL non-terminating at 26 dimensions	forced a different method	conjugation split, detection on about $N/2$ reals
maxcoeff 10^6 on a lift of height 6.3×10^6	reported $\dim K = 0$ at (16777216, 24)	construct known lifts rather than search for them
differences admitted where the rule was sums-only	inflated reachable set $4 \rightarrow 8$ of 10, understated a result by about $60\times$	restrict to the stated operation set

Pattern. Every forward claim in §14.1 and §14.2 — a quantity measured and reported — died. Every backward derivation — a constraint on what a relation must satisfy — survived, and two became derivations of the classical kernels rather than searches for them. Forward reasoning produces a witness and no information about what else exists; backward reasoning produces a condition on everything, so the count comes free. The single result in this paper that speaks to both halves of the operator, §7, is a backward one: it does not measure producers at all, it derives their uniqueness from a property of the annihilator.

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15. Open problems

1. **Ω — the 3-versus-5 asymmetry.** The octal seed admits a twisted annihilator selective for the 5-orbit and not the complementary 3-selective shape. The 3-orbit's prime content lives in $\mathbb{Q}(\sqrt{-2})$, the fixed field of σ_3 ; the 5-orbit's in $\mathbb{Q}(i)$, fixed by σ_5 — different decomposition groups. The twist that could annihilate the 3-orbit is indexed by the character of order 3, which requires $3 \mid q$ and puts one in $\mathbb{Q}(\zeta_{3n})$ where 3 ramifies and 5 does not. Candidate mechanism, not a theorem: the computation showing ramification forecloses the twisted relation has not been done.
2. **The remaining native dimension at (4096, 24).** One direction survives the closed integer-base ancestry. The four degree-4 irreducible factors of $x^{24} - 4096$ are not of the form $x^m - b_0$, so the next representation rung is algebraic-base ancestors, and that is where to aim.
3. **$\dim R = 7$ at (4096, 24).** Two dimensions above every reported measurement. Which reader attains them is unknown.
4. **Seed classification on the annihilator side.** Not “find more examples” but: characterise the complete set of primitive normalised generators. The negative half of the integer-base domain has never been swept and has already produced one seed of dimension 2.
5. **Complete the corrected table.** (16, 48), (16, 72), (256, 48), (256, 144), (4096, 36), (4096, 72) are unmeasured with the corrected machinery. Every dimension in the corpus came from a detector that was wrong by 2 at (16, 24) and by 2 in the other direction at (4096, 24). This is the widening move and should be run after §6 below, not before.
6. **When does a reader attain the ceiling?** §13 shows the question as posed is underdetermined: it names the coefficient field and not the twist field. The well-posed version asks for the joint condition on the pair, and the realisation map makes it a kernel computation rather than a search. This is the sharpest open problem the work produces.
7. **The exact generator of $\Sigma(K)$.** Divisibility by $U(b)$ is proved in §7.3. The measured generator is $U(b)$ or $3U(b)$ according to whether 9 divides $b - 1$, on seven frames. Deriving that dichotomy would make producer uniqueness computable from the factorisation of $b - 1$ alone, with no detection at any stage.
8. **Producer uniqueness at the confluence.** At (4096, 24) four dimensions survive C_1 . Whether a third independently motivated constraint closes them — and whether it is one the prior papers already contain — is open. The frame is the only measured failure of uniqueness among the eight published towers, and it is also the frame at which two lineages meet and at which $\dim R$ exceeds every reading.

Appendix A. Methods

A.1 Shell evaluation. S_r by direct summation to $\lceil 3.5 \text{ dps} / \log_{10}|b| \rceil + 80$ terms at dps 200–300. Negative bases converge as alternating series; no modification needed.

A.2 Relation lattice R . Complex conjugation splits the problem exactly. With $p_s = \text{Re } L_s$ (σ -symmetric) and $\zeta_s = \text{Im } L_s$ (antisymmetric), any c decomposes as $c^+ + c^-$ with the symmetric part carrying the real conditions and the antisymmetric part the imaginary. So $R = R^+ \oplus R^-$, each found by detection on about $N/2$ reals rather than a lattice reduction in $2N + 2$ dimensions. LLL at that size did not terminate.

A.3 Complete relation sets. Recursive elimination: find one relation by detection, drop the coordinate where the relation is largest, recurse. Complete, because any further relation reduces modulo the first to one supported off that coordinate. Residual gate 10^{-230} at dps 300; results stable from maxcoeff 10^5 to 10^7 . The gate is checked against the theoretical floor, not against zero, and a relation whose residual exceeds it is rejected rather than reported.

A.4 Kernel realisation over a named field. Given $w \in R$, $c_r = \lambda \hat{w}_r / (N\theta^r)$ with $\hat{w}_r = \sum_j w_j \omega^{(-jr)}$. Solvability of $C(o) = o$ is exactly $\sum_j w_j = o$ (Lemma 4.1). Membership of c_r in a named field is tested against that field's power basis. The phase λ must be free: $\lambda = 1$ for $w \in R^+$, $\lambda = i$ for $w \in R^-$. §13 shows that freeing λ is necessary but not sufficient — the R -side coefficient field must also be freed.

A.5 Uniqueness slack. For each frame, compute a complete integer basis of K by A.3, evaluate Σ on each basis vector, and report $\dim K$ minus the rank of Σ restricted to K . Rank is 1 whenever any basis Σ is non-zero and 0 otherwise, since Σ is a single functional. Where a known lift exists, construct it and check its residual before accepting a zero from the detector.

A.6 Precision discipline. Detection of a relation of length n with maximum coefficient G requires at least $n \log_{10} G$ working digits. Below that threshold the detector returns high-coefficient vectors that are artefacts of mantissa exhaustion rather than algebraic structure, and these are indistinguishable from results by inspection. Every dimension reported here was checked for stability across at least two coefficient bounds, and every residual against the theoretical floor for its frame.

Appendix B. Notes on scope

$\dim R$ is called frame-intrinsic on the strength of Baker's theorem, which fixes it against widening on the log side. That is a theorem. What is not established is that R as computed here — the $\mathbb{Q}$ -relation lattice among the principal-branch logs — is exhaustive of every analytic relation the frame supports. It is exhaustive of $\mathbb{Q}$ -linear ones. Claims beyond that are not made.

The sieve's completeness statement is bounded by its stated domain: $b \in [2, 320]$, $n \in [3, 9]$, positive bases. The negative half is unswept and has already produced a seed. No global classification is claimed anywhere in this paper.

The uniqueness result of §7 is a statement about the coefficient vector at a frame, not about the value a tower returns. Every value reported in Exactly Two Seeds §9 stands; what §7.2 adds is that at one of the eight frames the vector achieving it is not the only DC-blocked vector, and that this had not been checked. The uniqueness theorem of that paper's §10.1, which recovers the address (m, k) from the support, is a different statement and is untouched.

Little in this paper is unowned. The closed form and the valuation-vector method are Borwein–Borwein–Galway 2004. The formula for π is Bailey–Borwein–Plouffe 1997. Baker's theorem, Capelli's theorem, Zsigmondy and Bang, and the S-unit machinery are classical. What is offered as this work's is the direction of arrival and the specific reductions of §5, §6, §7 and §8. The claim that any of it is unstated in the literature rests on our own searching, which is not the same as a search, and should be checked properly before priority is asserted anywhere.

Appendix C. Code

Every numerical statement in this paper was produced by executed code before being written. The scripts below are the load-bearing ones and run standalone on mpmath, sympy and numpy.

C.1 Complete relation sets and uniqueness slack (§7). This is the script that produced the tables in §7.1 and §7.2.

```
import mpmath as mp
mp.mp.dps = 300

def shells(b, N):
    NT = int(3.5 * mp.mp.dps / mp.log10(abs(b))) + 120
    return [mp.fsum(mp.mpf(1) / ((N*k + r) * mp.mpf(b)**k)
        for k in range(NT)) for r in range(1, N)]

def kernel(S, maxcoeff=10**6):
    """complete integer relation set by recursive elimination (A.3)"""
    gate = mp.mpf(10) ** (-int(mp.mp.dps * 0.75))
    n = len(S); live = list(range(n)); out = []
    while len(live) > 1:
        sub = [S[i] for i in live]
        rel = mp.pslq(sub, maxcoeff=maxcoeff, maxsteps=400000,
            tol=mp.mpf(10) ** (-int(mp.mp.dps * 0.8)))
        if rel is None: break
        res = abs(mp.fsum(mp.mpf(c)*v for c, v in zip(rel, sub)))
```

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```

if res > gate: break          # gate against the floor, not zero
f = [0]*n
for c, i in zip(rel, live): f[i] = c
out.append((f, res))
live.pop(max(range(len(rel)), key=lambda k: abs(rel[k])))
return out

```

```

for b, N in [(16,8), (64,6), (16,12), (16,24), (4096,12),
            (4096,24), (-64,12), (256,16)]:
    K = kernel(shells(b, N))
    sums = [sum(f) for f, _ in K]
    rank = 1 if any(s != 0 for s in sums) else 0
    print((b, N), 'dimK', len(K), 'Sigma', sums,
          'slack', len(K) - rank)

```

C.2 Producer versus annihilator (§1.1). The three-line check that separates the two halves.

```

S8 = shells(16, 8)
print(mp.fsum(mp.mpf(c)*s for c, s in zip([8,-8,-4,-8,-2,-2,1], S8)))
# -> 9.68e-122    annihilator, Sigma = -15
print(mp.fsum(mp.mpf(c)*s for c, s in zip([4, 0, 0,-2,-1,-1,0], S8)))
# -> 3.14159265358979 producer, Sigma = 0

```

C.3 Divisibility of $\Sigma(K)$ by the unshared content (§7.3). Derived from the forcing lemma; the script only checks it.

```

from sympy import factorint, Poly, symbols, ZZ
from math import gcd
from functools import reduce

x = symbols('x')

def unshared_product(b):
    f = factorint(abs(b - 1))
    return reduce(lambda a, c: a*c,
                  [p for p, e in f.items() if e == 1 and p != 2], 1)

def orbit_norms(b, N):
    p = Poly(x**N - b, x, domain=ZZ)
    return [(str(g.as_expr()), int(g.eval(1)))]

```

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```

for g, _ in p.factor_list()[1]]

# for each frame: gcd of the measured Sigma values must be a multiple
# of unshared_product(b). 7/7 in the tested set.

```

C.4 Constructing a lift instead of searching for it (§7.2). The routine that caught the false negative at (16777216, 24).

```

D = {1: 16, 2: -24, 3: -8, 4: -6, 5: 1} # hex seed kernel
bo, m, q, k = 64, 6, 1, 4
W = {m*q*i: bo**(k-1-i) for i in range(k)} # 64^3 + 64^2 y^6 + 64 y^12 + y^18
C = {}
for wr, wc in W.items():
    for dr, dc in D.items():
        C[wr + dr*q] = C.get(wr + dr*q, 0) + wc*dc
# height 6291456, Sigma -5592405, residual 2.93e-295 against (16777216,24)

```

C.5 Reader reach as a kernel (§13). The derivation that reproduced two rungs and failed at the third. Retained because the failure is the result.

```

# c_r(w) = lambda * what_r(w) / (N theta^r) is LINEAR in w, so
# { w : all c_r(w) in F } is a Q-subspace and dim K_F is a nullity.
#
# what_r = sum_j w_j omega^(-j r)
# coordinates of each c_r taken in a basis of the ambient real field
# Q(z^(1/6), sqrt3) = { theta^i * sqrt3^j : i<6, j<2 }, degree 12
# reader F sees w iff every coordinate outside F's sub-basis vanishes;
# collect those as rows and take the rational nullity.
#
# RESULT: Q -> 1 (WRONG, measured 3), Q(theta) -> 3, Q(theta,sqrt3) -> 5
# The R-side was restricted to R (x) Q. The twisted classes live in
# R (x) Q(zeta_3), so two of the three directions were invisible.

```

Computational note. Every numerical statement in this paper was produced by executed code before being written. The producer/annihilator separation at (16, 8) and (64, 6); the complete relation sets and Σ values at eight frames at dps 300 with residuals from 7.9×10^{-301} to 1.5×10^{-297} ; the eight-tower uniqueness table; the constructed $k = 4$ hexagonal lift at height 6291456 with residual 2.93×10^{-295} ; the Σ -divisibility check at 7/7 with orbit norms from exact factorisation; $\dim K(16, 24) = 3$ with class-pure supports and heights 8, 32, 48, stable across maxcoeff $10^5, 10^6, 10^7$ at residuals of order 10^{-301} ; $\dim R = 1, 1, 1, 5$ with the R^+/R^- split; the reader-reach nullities 1, 3, 5 and

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the ζ_3 diagnosis of the failure at $\mathbb{Q}$; and the lift heights 128, 2048, 1536, 6291456. Python with mpmath, sympy and numpy at 120–300 decimal digits as precision required. Source accompanies this paper.

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